

ROOT CAUSE ANALYSIS

Clarify (Getting on the same page)

Problem Statement: As a Product Manager at Netflix, we've observed a 10% drop in Average View Duration (AVD). We need to identify the root cause.

Defining the Metric: Average View Duration (AVD) refers to the average amount of time a user spends watching content on Netflix. It measures *how long* users are watching, not *how many* users are watching. For example, if last week users were watching an average of 45 minutes per day and this week it dropped to approximately 40.5 minutes per day, that represents a ~10% decline.

Important distinction: AVD is not the same as active user count or viewer count. The same number of users could be logging in, but spending less time watching that's an AVD drop.

Scope & Timeframe: We are investigating a 10% week-over-week drop in average daily viewing time per user on Netflix.

Basic Sanity Checks

Before assuming the drop is a real behavioral change, we need to rule out measurement and contextual issues:

1. **Data pipeline and tooling accuracy** - Has the analytics team verified that the data collection, processing pipeline, and reporting tools are functioning correctly? A 10% drop is significant and warrants double verification.
2. **Metric definition change** - Has the internal definition of AVD been recently modified? For example, if mobile previews or trailers were recently excluded from the AVD calculation, that alone could explain the drop without any actual change in user behavior.
3. **A/B tests or feature rollouts** - Were there any recent experiments or feature deployments that may have affected how viewing time is logged or attributed? Experiments that split traffic can sometimes distort aggregate metrics.
4. **Seasonality or calendar effects** - Was the prior week's baseline inflated due to holidays, school breaks, or long weekends? The current "drop" could simply be a return to normal levels rather than an actual decline.

Analyzing What to Target First

A) Industry-Wide or Netflix-Specific?

Before diving into root causes, we need to determine whether this 10% drop in AVD is a Netflix-specific problem or part of a broader industry-wide decline in streaming viewership.

How to assess this:

1. **Public earnings reports and investor calls** - Review quarterly reports from competing streaming platforms (Disney+, Amazon Prime Video, Warner Bros Discovery) for any reported drops in engagement or viewing hours.
2. **Third-party analytics firms** - Leverage data from Nielsen, Sensor Tower, App Annie, and Comscore, which track cross-platform streaming viewership and engagement trends.
3. **Industry news and trade publications** - Monitor sources like Variety, The Verge, and TechCrunch for any reports on broader streaming viewership declines.
4. **App store and social media signals** - Check whether competitors are experiencing drops in app downloads, daily active usage, or rising user complaints about content quality.
5. **Google Trends** - Compare search interest across major streaming platforms to identify whether overall streaming interest is declining or if the dip is isolated to Netflix.

Conclusion: After conducting the above research, we've determined that the drop is **Netflix-specific**. Competing platforms are holding steady or showing growth during the same period. This tells us the root cause lies within Netflix's own ecosystem.

Segmenting the Metric to Narrow the Problem

Before investigating causes, we need to segment the AVD drop to understand *where* and *for whom* it is happening. This helps narrow the investigation significantly.

1. **Platform** - Is the drop occurring on Smart TVs, mobile devices, tablets, web browsers, or uniformly across all platforms? For example, a drop isolated to Smart TVs could point to a firmware or app update issue.
2. **Geography** - Is this a global decline or concentrated in specific countries or regions? A drop isolated to a specific market could point to content licensing changes, regulatory issues, or localized technical problems in that region.
3. **User segments** - Are new users, long-term subscribers, or free trial users disproportionately affected? A drop among new users might suggest an onboarding or content discovery problem, while a drop among long-term users could indicate content fatigue.
4. **Content type** - Is the drop spread across movies, series, documentaries, and originals equally? Or is it concentrated in a specific content category? For instance, if series AVD dropped but movie AVD stayed flat, it could indicate a weak slate of recent series releases.

5. **Time of day** - Is the drop happening during prime-time evening hours, daytime, or uniformly throughout the day? A prime-time-specific drop could indicate competition from live events or broadcast TV.
6. **Subscription tier** - Are Basic, Standard, or Premium tier users equally affected? A drop concentrated in the ad-supported Basic tier might suggest that ad frequency or ad experience is driving users away mid-session.

Conclusion from Segmentation:

After analyzing the data across all segments, we've found that the 10% AVD drop is **Uniform Across The Board**, all platforms, all geographies, all user segments, all content types, and all subscription tiers are equally affected.

External Factors (Things Not Under Netflix's Control)

Since we've established that this is a Netflix-specific drop, we focus only on external factors that would disproportionately impact Netflix while leaving competitors relatively unaffected.

1. **Competitor launching a more engaging feature** - Other Streaming platforms (Disney+, Prime Video, etc.) may have introduced a highly engaging feature such as live sports streaming, interactive content, or a social co-watching experience that is pulling Netflix users to spend more of their viewing time on that platform instead.
2. **Major exclusive content launch on a competitor** - A blockbuster release on a competing platform (e.g., a massive Marvel series on Disney+ or a highly anticipated show on Prime Video) could temporarily divert Netflix users' attention and reduce their daily viewing time on Netflix.
3. **Negative publicity or controversy specific to Netflix** - Netflix has historically faced backlash over content decisions, pricing hikes, and policy changes like password-sharing crackdowns. Any recent controversy or negative media coverage could lead to users consciously or subconsciously reducing their engagement with the platform.
4. **Regulatory or policy changes targeting Netflix** - A government in a key market may have banned specific Netflix content, imposed new content regulations, or introduced data/streaming restrictions that reduced the available content library or degraded the user experience in that region.

After investigating, external factors are **unlikely to be the primary driver** of this drop. Here's why:

- A competitor feature launch or blockbuster content release would typically pull users away in specific demographics or regions but our segmentation shows the drop is **uniform across all geographies and user segments**, which doesn't fit this pattern.

- Regulatory or policy changes are inherently regional but the drop is **global**, ruling this out as a primary cause.
- Negative publicity or controversy remains a possibility, but significant controversy would typically show stronger impact in the market where the news originated and among specific demographics more tuned into media coverage. A perfectly uniform global drop doesn't align well with this either.

Since external factors don't convincingly explain a systemic, uniform, global drop, the root cause most likely lies within Netflix's internal ecosystem a product-wide change that affects every user equally.

Internal Factors (Things Not Under Netflix's Control)

Since external factors have been largely ruled out, and the drop is systemic, uniform, and global across all segments, the root cause most likely lies within Netflix's internal ecosystem. We investigate across four key categories: Content, Product/UX, Technology, and Business Decisions.

A. Content

1. **Weak slate of new releases:** If Netflix's recent content pipeline has been underwhelming no breakout original series, no highly anticipated new seasons, or flagship shows ending without strong replacements users may still log in but find nothing compelling enough to keep them watching for extended periods. They start titles, don't get hooked, and abandon sooner, directly reducing AVD.
2. **Content quality decline:** Even if the volume of new releases is consistent, a drop in overall quality means users are less likely to binge. They might watch one episode instead of five, or abandon a movie halfway through. Same number of users showing up, but significantly less time spent watching.

B. Product / UX

3. **Recommendation algorithm update:** Netflix's recommendation engine is the primary driver of content discovery and engagement. If a recent algorithm update is surfacing less relevant or less personalized content, users end up watching titles that don't match their preferences. This leads to faster abandonment, more title-switching, and shorter overall viewing sessions — directly reducing AVD even though users are still showing up.
4. **Autoplay and binge mechanics changes:** This is one of the strongest candidates for a uniform global drop. Netflix's autoplay next episode feature is a

critical driver of extended viewing sessions. Any of the following changes could significantly reduce AVD:

- Increasing the countdown timer between episodes (e.g., from 5 seconds to 15 seconds), giving users more opportunity to disengage.
 - Increasing the frequency of "Are you still watching?" prompts (e.g., from every 4 episodes to every 2 episodes), breaking binge momentum.
 - Turning off autoplay by default due to a regulatory requirement or UX decision. These changes would impact every user, on every platform, in every geography perfectly matching the uniform pattern observed in our segmentation.
5. **UI/UX redesign:** If Netflix recently rolled out a global UI change such as modified navigation, altered content browsing layouts, or changes to how users interact with playback controls this could disrupt familiar user behavior and reduce the ease of discovering and continuing content, leading to shorter viewing sessions.

C. Technology

6. **Audio/video quality or playback issues:** Degradation in streaming quality such as buffering, resolution drops, audio sync issues, or increased loading times would directly frustrate users mid-viewing and cause them to stop watching sooner. Even subtle quality degradation can reduce how long users are willing to keep watching.
7. **Technical bugs causing interruptions:** Bugs such as unexpected app crashes, playback freezing, subtitle malfunctions, or episodes not loading correctly would interrupt the viewing experience and directly reduce AVD.

D. Business Decisions

8. **Introduction or increase of ad interruptions:** If Netflix recently introduced ads on previously ad-free tiers, increased ad frequency on the ad-supported tier, or changed ad placement (e.g., mid-roll ads instead of pre-roll), this would directly interrupt viewing sessions. Users experiencing frequent ad breaks are more likely to stop watching sooner, reducing AVD.

Conclusion from Internal Factors:

Given that the drop is perfectly uniform across all platforms, geographies, user segments, content types, and subscription tiers, the root cause is most likely a **systemic product-wide change** rather than a content or technology issue.

- Content issues (weak releases, quality decline) would typically show up in specific content categories first, not uniformly across all content types.

- Technology issues (bugs, quality degradation) would typically be more pronounced on specific platforms or regions, not perfectly uniform everywhere.
- Business decisions like ad changes would primarily affect specific subscription tiers, not all tiers equally.

The most probable root causes are **product/UX changes that apply globally to every user** specifically:

- **A recommendation algorithm update that is surfacing less engaging content across the board.**
- **A change in autoplay or binge mechanics that is breaking viewing momentum for every user equally.**

These are the highest-priority areas to investigate first, as they are the only internal changes that would logically produce a perfectly uniform, global, cross-segment drop in Average View Duration.

USER JOURNEY

To pinpoint exactly where in the Netflix experience the AVD drop is occurring, we map the complete user journey from trigger to exit and identify the critical drop-off zones.

Triggers (Why does the user open Netflix?)

Users come to Netflix driven by a variety of motivations:

1. They are bored and looking for entertainment to pass time.
2. They want to relax after a long working day.
3. They want to take a break from work or daily routine.
4. They have a specific movie or show in mind that they want to watch.
5. They want to have fun with family or friends by watching something together.
6. A new release from their favourite actor, director, or a franchise they follow has arrived on the platform.

These triggers remain unchanged users are still showing up to Netflix. The problem is not with motivation to visit, it's with what happens after they arrive.

User Actions (What does the user do inside Netflix?)

Once the user opens Netflix and logs in, their journey follows these steps:

1. **Home Page / Browse:** The user lands on the home page where Netflix's recommendation algorithm surfaces content through sections like "For You," "Trending in Your Country," "Trending Globally," "New Releases," and "Continue Watching."
2. **Search (if needed):** If the user doesn't find anything appealing on the home page, they use the search function to look for a specific title by name, genre, or actor.
3. **Title Evaluation:** The user evaluates a potential title by reading the description, checking ratings, viewing the Netflix match score (how likely they are to enjoy it), and watching the preview/trailer.
4. **Press Play:** The user selects a title and begins watching.
5. **During Viewing:** The user is actively watching the content. Their engagement depends on how compelling the content is, the quality of audio/video, and whether there are any interruptions (ads, buffering, bugs).
6. **Episode Transition:** When an episode ends, the autoplay countdown begins. The next episode either starts automatically, or the user is prompted with "Are you still watching?" This is the critical decision point that determines whether the user continues binging or stops.
7. **Repeat or Browse Again:** If the user finishes a movie or series, they may browse for something new, or they may decide they've watched enough and exit.

Exit Points (Why does the user stop watching?)

Users stop watching and leave Netflix for the following reasons:

1. External interruptions like work, phone calls, responsibilities, sleep.
2. They got bored with what they were watching.
3. They couldn't find anything interesting to watch in the first place.
4. Technical issues buffering, crashes, audio/video problems.
5. Ad interruptions reduced their enjoyment.
6. The autoplay transition gave them enough time or a prompt to consciously decide to stop.

Mapping the Three Critical Drop-Off Zones

Based on our internal factors analysis, we identify three specific zones in the user journey where AVD is most likely dropping:

Drop-Off Zone 1 – Discovery Phase (Steps 1-3)

If Netflix's recommendation algorithm was recently updated and is now surfacing less relevant or less personalized content, users struggle to find something they want to

watch. They browse the home page, scroll through categories, maybe try a search but nothing grabs them. They either leave quickly without watching anything, or settle for something they're not excited about (which leads to early abandonment during viewing). This reduces the total time spent watching per day.

Drop-Off Zone 2 – During Viewing (Step 5)

Even after pressing play, users may abandon content midway if the content quality isn't strong enough to hold attention, if there are audio/video quality issues, or if ad interruptions break the viewing experience. A user who would have watched a full 2-hour movie but stops at the 40-minute mark because the content didn't hook them directly reduces AVD.

Drop-Off Zone 3 – Episode Transition (Step 6)

This is the most critical zone and the most likely primary driver of the uniform global AVD drop. Here's why:

The episode transition is the moment between episodes where the user either continues watching or stops. Netflix's autoplay feature has historically been the biggest driver of extended viewing sessions the next episode starts automatically within seconds, keeping users in a continuous viewing loop.

If Netflix made any changes to this mechanic increasing the autoplay countdown timer, showing "Are you still watching?" prompts more frequently, or modifying autoplay defaults the impact compounds across every session:

- A typical binge session involves 3-4 episode transitions.
- Even a small reduction in continuation rate at each transition (e.g., from 90% to 83%) compounds significantly: $0.9^3 = 73\%$ still watching after 3 transitions vs $0.83^3 = 57\%$. This means average sessions shrink from ~4 episodes to ~3 episodes a substantial AVD reduction.
- This change applies identically to every user, every platform, every geography, every content type, and every subscription tier perfectly explaining the uniform, global pattern observed in our segmentation.

No other factor produces such a perfectly uniform impact across all segments.

Conclusion from User Journey Analysis:

The 10% AVD drop is most likely driven by a combination of factors, with the episode transition mechanics change as the primary driver, supported by secondary

contributions from the discovery phase (recommendation algorithm serving less relevant content) and the viewing phase (content quality not strong enough to retain attention).

The episode transition is identified as the primary driver because it is the only factor that logically produces a perfectly uniform drop across all platforms, geographies, user segments, content types, and subscription tiers which is exactly the pattern confirmed in our segmentation analysis.

Priority investigation order:

1. **First:** Audit any recent changes to autoplay countdown timers, "Are you still watching?" prompt frequency, and autoplay default settings.
2. **Second:** Evaluate recent recommendation algorithm updates and their impact on content relevance scores.
3. **Third:** Assess the current content slate quality and release pipeline against previous quarters.

SUMMARY

Root Cause: The 10% week-over-week drop in Average View Duration is a Netflix-specific, systemic issue that is uniform across all platforms, geographies, user segments, content types, and subscription tiers. After ruling out external factors, data/tooling issues, and seasonality, the most probable root cause is a **recent change in autoplay and binge mechanics** (increased countdown timers, more frequent "Are you still watching?" prompts), with secondary contributions from a **recommendation algorithm update** and a **weaker content slate**.

Short-Term Fixes:

1. Roll back any recent autoplay/binge mechanic changes and monitor AVD recovery.
2. A/B test autoplay configurations to quantify exact impact.
3. Audit and roll back the recommendation algorithm to the last stable version if degradation is confirmed.

Long-Term Solutions:

1. Build personalized autoplay tuning based on individual user binge patterns.

2. Strengthen the recommendation engine with real-time engagement signals.
3. Ensure a consistent cadence of high-quality content releases to avoid content droughts.

Key Metrics to Monitor: Average View Duration, episode continuation rate, content completion rate, and time to first play.

Key Takeaway: The fastest path to recovery is auditing and reverting recent product-wide changes to autoplay and recommendation systems. The uniform nature of the drop strongly suggests a mechanical product change rather than a content or external market issue.